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STATOR FOR ELECTRIC MACHINE

Abstract

A stator for an electric machine includes a stator core defining a slot extending in an axial direction from a first end to a second end, a wire disposed in the slot, a coolant pipe disposed in the slot and extending in the axial direction from the first end to the second end, the coolant pipe extending from a pipe inlet to a pipe outlet, and a thermally conductive potting material disposed in the slot between the coolant pipe, the wire, and the stator core.

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Background/Summary

FIELD

[0001] The present disclosure relates to gas turbine engines, and more specifically to cooling

devices in gas turbine engines.

BACKGROUND

[0002] Typical aircraft propulsion systems include one or more gas turbine engines. For certain propulsion systems, the gas turbine engines generally include a fan and a core arranged in flow communication with one another. Additionally, the core of the gas turbine engine generally includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. In operation, air is provided from the fan to an inlet of the compressor section where one or more axial compressors progressively compress the air until it reaches the combustion section. Fuel is mixed with the compressed air and burned within the combustion section to provide combustion gases. The combustion gases are routed from the combustion section to the turbine section. The flow of combustion gasses through the turbine section drives the turbine section and is then routed through the exhaust section, e.g., to atmosphere.

[0003] Incorporating an electrical machine (e.g., an electrical generator) into a propulsion engine to generate electrical power from mechanical energy generated by the propulsion engine may enhance the capabilities of aircraft. For example, the electrical power generated by the electrical machine may be used to operate an accessory propulsor (e.g., an electric fan, motor, or the like) to supplement thrust provided via the turbine engine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0005] FIG. 1 is a cross-sectional view of an exemplary gas turbine engine.

[0006] FIG. 2 is a perspective view of an exemplary stator of an electric machine of the exemplary gas turbine engine of FIG. 1.

[0007] FIG. 3 is a partial magnified view of the stator of FIG. 2 illustrating coolant pipes.

[0008] FIG. 4 is a partial magnified view of the stator of FIG. 2 illustrating coolant flow through the coolant pipes of FIG. 3.

[0009] FIG. 5 is a perspective view of another exemplary stator.

[0010] FIG. 6 is a partial magnified view of the stator of FIG. 5 illustrating coolant pipes.

[0011] FIG. 7 is a partial magnified view of the stator of FIG. 2 illustrating coolant flow through the coolant pipes of FIG. 6.

[0012] FIG. 8 is a block diagram of an exemplary method for forming an electric machine.

DETAILED DESCRIPTION

[0013] Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

[0014] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

[0015] The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

[0016] The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which fluid flows,

and “downstream” refers to the direction to which the fluid flows.

[0017] The present disclosure is generally related to devices for cooling electric machines in gas turbine engines, as well as methods of forming such devices. During operation, electric machines (such as electric motors and electric generators) generate heat. The heat may interfere with operation of the electric machine, such as increasing an electrical resistance of wires that leads to a decreased magnetic field generated by the wires. Thermal management to dissipate generated heat leads to improved efficiency and lifespan of the electric machines.

[0018] Liquid coolants can provide heat transfer by convection, and solid thermal sinks can provide heat transfer by conduction. Providing a liquid coolant in pipes in the stator dissipates heat from the wires. Applying a potting material that has a high thermal conductivity to ends of the stator further dissipates heat from the wires. By combining the convective heat transfer of the coolant and the conductive heat transfer of the potting material, the overall heat dissipation from the wires increases, and the efficiency of the electric machine is improved.

[0019] Referring now to FIG. 1, a schematic cross-sectional view of a gas turbine engine according to one example embodiment of the present disclosure is shown. Particularly, FIG. 1 provides an aviation three-stream turbfan engine herein referred to as “three-stream engine **100**”. The three-stream engine **100** of FIG. 1 can be mounted to an aerial vehicle, such as a fixed-wing aircraft, and can produce thrust for propulsion of the aerial vehicle. The three-stream engine **100** is a “three-stream engine” in that its architecture provides three distinct streams of thrust-producing airflow during operation.

[0020] For reference, the three-stream engine **100** defines an axial direction A, a radial direction R, and a circumferential direction C. Moreover, the three-stream engine **100** defines an axial centerline or longitudinal axis **112** that extends along the axial direction A. In general, the axial direction A extends parallel to the longitudinal axis **112**, the radial direction R extends outward from and inward to the longitudinal axis **112** in a direction orthogonal to the axial direction A, and the circumferential direction extends three hundred sixty degrees (360°) around the longitudinal axis **112**. The three-stream engine **100** extends between a forward end **114** and an aft end **116**, e.g., along the axial direction A.

[0021] The three-stream engine **100** includes a core engine **118** and a fan section **150** positioned upstream thereof. Generally, the core engine **118** includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. Particularly, as shown in FIG. 1, the core engine **118** includes an engine core **120** and a core cowl **122** that annularly surrounds the engine core **120**. The engine core **120** and core cowl **122** define an annular core inlet **124**. The core cowl **122** further encloses and supports a booster or low pressure compressor **126** for pressurizing the air that enters the core engine **118** through core inlet **124**. A high pressure, multi-stage, axial-flow compressor (hereinafter, a high pressure compressor **128**) receives pressurized air from the LP compressor **126** and further increases the pressure of the air. The pressurized air stream flows downstream to a combustor **130** where fuel is injected into the pressurized air stream and ignited to raise the temperature and energy level of the pressurized air.

[0022] The high energy combustion products flow from the combustor **130** downstream to a high pressure turbine **132**. The high pressure turbine **132** drives the high pressure compressor **128** through a first shaft or high pressure shaft **136**. In this regard, the high pressure turbine **132** is drivingly coupled with the high pressure compressor **128**. The high energy combustion products then flow to a low pressure turbine **134**. The low pressure turbine **134** drives the low pressure compressor **126**, components of the fan section **150**, and an electric machine **200** through a second shaft or low pressure shaft **138**. Specifically, the high energy combustion products drive turbine blades **135** of the low pressure turbine **134**. In this regard, the low pressure turbine **134** is drivingly coupled with the low pressure compressor **126**, components of the fan section **150**, and the electric machine **200**. The LP shaft **138** is coaxial with the HP shaft **136** in this example embodiment. After driving each of the turbines **132**, **134**, the combustion products exit the core engine **118** through a

core exhaust nozzle **140** to produce propulsive thrust. Accordingly, the core engine **118** defines a core flowpath or core duct **142** that extends between the core inlet **124** and the core exhaust nozzle **140**. The core duct **142** is an annular duct positioned generally inward of the core cowl **122** along the radial direction R.

[0023] The fan section **150** includes a primary fan **152**. For the depicted embodiment of FIG. **1**, the primary fan **152** is an open rotor or unducted primary fan **152**. However, in other embodiments, the primary fan **152** may be ducted, e.g., by a fan casing or nacelle circumferentially surrounding the primary fan **152**. As depicted, the primary fan **152** includes an array of fan blades **154** (only one shown in FIG. **1**). The fan blades **154** are rotatable, e.g., about the longitudinal axis **112**. As noted above, the primary fan **152** is drivably coupled with the low pressure turbine **134** via the LP shaft **138**. The primary fan **152** can be directly coupled with the LP shaft **138**, e.g., in a direct-drive configuration. Optionally, as shown in FIG. **1**, the primary fan **152** can be coupled with the LP shaft **138** via a speed reduction gearbox **155**, e.g., in an indirect-drive or geared-drive configuration.

[0024] Moreover, the fan blades **154** can be arranged in equal spacing around the longitudinal axis **112**. Each of the fan blades **154** has a root and a tip and a span defined therebetween. Each of the fan blades **154** defines a central blade axis **156**. For this embodiment, each of the fan blades **154** of the primary fan **152** is rotatable about their respective central blades axes **156**, e.g., in unison with one another. One or more actuators **158** can be controlled to pitch the fan blades **154** about their respective central blades axes **156**. However, in other embodiments, each of the fan blades **154** may be fixed or unable to be pitched about its central blade axis **156**.

[0025] The fan section **150** further includes a fan guide vane array **160** that includes fan guide vanes **162** (only one shown in FIG. **1**) disposed around the longitudinal axis **112**. For this embodiment, the fan guide vanes **162** are not rotatable about the longitudinal axis **112**. Each fan guide vane **162** has a root and a tip and a span defined therebetween. The fan guide vanes **162** may be unshrouded as shown in FIG. **1** or may be shrouded, e.g., by an annular shroud spaced outward from the tips of the fan guide vanes **162** along the radial direction R. Each fan guide vane **162** defines a central blade axis **164**. For this embodiment, each fan guide vane **162** of the fan guide vane array **160** is rotatable about their respective central blades axes **164**, e.g., in unison with one another. One or more actuators **166** can be controlled to pitch the fan guide vane **162** about their respective central blades axes **164**. However, in other embodiments, each fan guide vane **162** may be fixed or unable to be pitched about its central blade axis **164**. The fan guide vanes **162** are mounted to a fan cowl **170**.

[0026] The fan cowl **170** annularly encases at least a portion of the core cowl **122** and is generally positioned outward of the core cowl **122** along the radial direction R. Particularly, a downstream section of the fan cowl **170** extends over a forward portion of the core cowl **122** to define a fan flowpath or fan duct **172**. Incoming air may enter through the fan duct **172** through a fan duct inlet **176** and may exit through a fan exhaust nozzle **178** to produce propulsive thrust. The fan duct **172** is an annular duct positioned generally outward of the core duct **142** along the radial direction R. The fan cowl **170** and the core cowl **122** are connected together and supported by a plurality of substantially radially-extending, circumferentially-spaced struts **174** (only one shown in FIG. **1**). The struts **174** may each be aerodynamically contoured to direct air flowing thereby. Other struts in addition to struts **174** may be used to connect and support the fan cowl **170** and/or core cowl **122**.

[0027] The three-stream engine **100** also defines or includes an inlet duct **180**. The inlet duct **180** extends between an engine inlet **182** and the core inlet **124**/fan duct inlet **176**. The engine inlet **182** is defined generally at the forward end of the fan cowl **170** and is positioned between the primary fan **152** and the fan guide vanes **162** along the axial direction A. The inlet duct **180** is an annular duct that is positioned inward of the fan cowl **170** along the radial direction R. Air flowing downstream along the inlet duct **180** is split, not necessarily evenly, into the core duct **142** and the fan duct **172** by a nose of a splitter **144** of the core cowl **122**. The inlet duct **180** is wider than the core duct **142** along the radial direction R. The inlet duct **180** is also wider than the fan duct **172**

along the radial direction R.

[0028] As depicted, the fan section **150** also includes a mid-fan **190**. The mid-fan **190** includes an array of mid-fan blades **192** (only one shown in FIG. 1). The mid-fan blades **192** are rotatable, e.g., about the longitudinal axis **112**. The mid-fan **190** is drivably coupled with the low pressure turbine **134** via the LP shaft **138**. The mid-fan blades **192** can be arranged in equal circumferential spacing around the longitudinal axis **112**. Each mid-fan blade **192** has a root **194** and a tip **196** and a span defined therebetween. Moreover, each mid-fan blade **192** has a leading edge **198** and a trailing edge **199**. The mid-fan blades **192** are annularly surrounded or ducted by the fan cowl **170**. In this regard, the mid-fan **190** is positioned inward of the fan cowl **170** along the radial direction R. Moreover, for this example embodiment, the mid-fan **190** is positioned within the inlet duct **180** upstream of both the core duct **142** and the fan duct **172**.

[0029] Accordingly, air flowing through the inlet duct **180** flows across the mid-fan blades **192** and is accelerated downstream thereof, particularly at the tips **196** of the mid-fan blades **192**. At least a portion of the air accelerated by the mid-fan blades **192** flows into the fan duct **172** and is ultimately exhausted through the fan exhaust nozzle **178** to produce propulsive thrust. Also, at least a portion of the air accelerated by the mid-fan blades **192** flows into the core duct **142** and is ultimately exhausted through the core exhaust nozzle **140** to produce propulsive thrust. Generally, the mid-fan **190** is a compression device positioned downstream of the engine inlet **182**. The mid-fan **190** is operable to accelerate air into the fan duct **172** or secondary bypass passage.

[0030] It will be appreciated, however, that the three-stream engine **100** is provided by way of example only. In other exemplary embodiments, the three-stream engine **100** may have any other configuration. For example, in other exemplary embodiments, the engine core **120** may have any other number and arrangement of shafts, spools, compressors, turbines, etc. Further, in other exemplary embodiments, the three-stream engine **100** may alternatively be configured as a ducted turbofan engine (including an outer nacelle surrounding the primary fan **152** and a portion of the engine core **120**); as a direct drive gas turbine engine (may not include a reduction gearbox, such as the speed reduction gearbox **155**); as a fixed pitch gas turbine engine (may not include a variable pitch fan, such as the primary fan **152**); as a two-stream gas turbine engine (may not include the fan duct **172**); etc.

[0031] Further, for the depicted embodiment of FIG. 1, the three-stream engine **100** includes an electric machine operably coupled with a rotating component thereof. In this regard, the three-stream engine **100** is an aeronautical hybrid-electric propulsion machine. Particularly, as shown in FIG. 1, the three-stream engine **100** includes electric machine **200** operatively coupled with the LP shaft **138**. The electric machine **200** includes a rotor **202** and a stator **204**. The electric machine **200** can be directly mechanically connected to the LP shaft **138**, or alternatively, the electric machine **200** can be mechanically coupled with the LP shaft **138** indirectly, e.g., by way of a gearbox. Further, although the electric machine **200** is operatively coupled with the LP shaft **138** at an aft end of the LP shaft **138**, the electric machine **200** can be coupled with the LP shaft **138** at any suitable location or can be coupled to other rotating components of the three-stream engine **100**, such as the HP shaft **136**.

[0032] In some embodiments, the electric machine **200** can be an electric motor operable to drive or motor the LP shaft **138**, e.g., during an engine burst. In other embodiments, the electric machine **200** can be an electric generator operable to convert mechanical energy into electrical energy. In this way, electrical power generated by the electric machine **200** can be directed to various engine and/or aircraft systems. In some embodiments, the electric machine **200** can be a motor/generator with dual functionality.

[0033] Now referring to FIG. 2, an exemplary stator **204** for an electric machine **200** (specifically, an electric motor) is shown. The stator **204** defines an axial direction A', a radial direction R', and a circumferential direction C'. In certain exemplary embodiments, the exemplary stator **204** of FIG. 2 may be incorporated into the electric machine **200** of FIG. 1. In such an exemplary embodiment, it

will be appreciated that the axes used for the directions A', R', and C' are defined relative to the electric machine **200** and may or may not align with the directions A, R, C for the three-stream engine **100**.

[0034] The stator **204** includes a stator core **206** defining a plurality of slots **208** extending in the axial direction A' from a first end **210** of the stator core **206** to a second end **212** of the stator core **206**. The stator **204** includes a plurality of wires **214** arranged as windings disposed in the slots **208** and a plurality of coolant pipes **216** disposed through the plurality of slots **208**, the plurality of coolant pipes **216** extending in the axial direction A' from the first end **210** to the second end **212**. Each of the plurality of coolant pipes **216** extends from a pipe inlet **218** to a pipe outlet **220** (best shown in FIGS. 4-4B). The stator core **206** includes a manifold **222** that extends around the stator core **206** in the circumferential direction C', and the manifold **222** fluidly connects to each of the plurality of coolant pipes **216**. The stator **204** includes a coolant supply **224** that fluidly connects to the manifold **222** to supply a coolant **226** to the manifold **222** and to the coolant pipes **216**. The stator **204** includes a thermally conductive potting material **228** disposed in the plurality of slots **208** between the plurality of coolant pipes **216**, the plurality of wires **214**, and the stator core **206**.

[0035] The stator core **206** is the stationary part of the electric machine **200**, providing structure to house the rotor **202** (not shown) that rotates within the stator **204**. The stator core **206** and the plurality of wires **214** act as an electromagnet during operation of the electric machine **200**. That is, when the rotor **202** is rotating within the stator **204**, the rotor **202** induces electric current through the plurality of wires **214**, generating electric current. In such a configuration, the electric machine **200** acts as a generator to generate electricity for use by other components.

[0036] Alternatively, by providing electric current to the plurality of wires **214**, the stator **204** generates a magnetic field that rotates the rotor **202**, providing rotational motion to a rotor shaft (not shown) to drive other components. In such a configuration, the electric machine **200** acts as a motor that drives rotation of another component.

[0037] During operation of the electric machine, electrical resistance in the plurality of wires **214** generates heat that may interfere with electric current generation. As will be explained in greater detail below, the plurality of coolant pipes **216** and the thermally conductive potting material **228** dissipate heat from the plurality of wires **214**, improving performance of the electric machine **200**.

[0038] As shown in FIG. 2, the stator **204** includes the thermally conductive potting material **228**. The potting material **228** is disposed on the first end **210** of the stator core **206** and the second end **212** of the stator core **206**, extending into the plurality of slots **208** between the plurality of coolant pipes **216**, the plurality of wires **214**, and the stator core **206**, covering the first and second ends **210**, **212** and the manifold **222**. The potting material **228** absorbs heat from the plurality of coolant pipes **216** and the plurality of wires **214**, increasing heat transfer out from the stator **204**. The potting material **228** may include an electrically-insulative material, such as an epoxy resin, a silicon-based resin, or both. The potting material **228** is applied as a liquid or gel, and then the potting material **228** is cured to a solid in a heat treatment or drying process. The solid potting material **228**, being in contact with the stator core **206** and the plurality of wires **214**, fixes the plurality of wires **214** within the plurality of slots **208**. The potting material **228** thus secures the plurality of wires **214** to the stator core **206**, improving structural integrity and conductive heat transfer of the stator **204**.

[0039] The stator core **206** includes a plurality of wedges **230**, each wedge **230** enclosing the respective wire **214** and the respective coolant pipe **216** within the respective slot **208**. The wedges **230** are angular-shaped components that are shaped to fixedly attach to the stator core **206**, held in place in the slot **208** by friction. The plurality of wedges **230** and the stator core **206** form an inner surface **232** of the stator **204** in the radial direction R'. The plurality of wedges **230** may be substantially flush with the stator core **206** such that the inner surface **232** is substantially smooth.

[0040] Now referring to FIG. 3, a close-up view of the stator **204** of FIG. 2 is shown with the potting material **228** removed and the plurality of wires **214** shown in phantom lines to illustrate

the arrangement of the coolant pipes **216** and the wires **214** within the slots **208**. As described above, the plurality of coolant pipes **216** provide the coolant **226** to the plurality of wires **214**, transferring the heat generated by the wires **214** away from the stator **204**. Because the plurality of wires **214** extend from the first end **210** to the second end **212**, heat is generated along the entire axial length of the slots **208**. Accordingly, the plurality of coolant pipes **216** extend through the plurality of slots **208** from the first end **210** to the second end **212** to dissipate heat from the entire length of the plurality of wires **214**. In order to dissipate the heat from the plurality of wires **214**, at least one of the plurality of wires **214** transfers heat to the coolant pipe **216** via conduction, either by direct contact or with an intermediary heat transfer medium. The coolant **226** flowing through the coolant pipe **216** is heated by contact with the coolant pipe **216**, and the heated coolant **226** flows out of the slot **208**, dissipating heat out from the stator **204**.

[0041] Now referring to FIG. **4**, the stator **204** of FIG. **3** is shown with the stator core **206** and the plurality of wires **214** removed to illustrate the manifold **222** and the coolant pipes **216**. In the exemplary embodiment of FIG. **4**, the pipe inlet **218** of each coolant pipe **216** is disposed at the first end **210** and the pipe outlet **220** of each coolant pipe **216** is disposed at the second end **212**. The coolant pipes **216** are arranged in parallel to each other to allow the coolant **226** to flow through each of the coolant pipes **216** substantially simultaneously.

[0042] The manifold **222** extends around the stator core **206** in the circumferential direction C' outward of the stator core **206** in the radial direction R', fluidly connecting to each of the plurality of coolant pipes **216**. The manifold **222** of FIG. **4** includes a first portion **234** including a manifold inlet **236** and a second portion **238** including a manifold outlet **240**. The first portion **234** is disposed at the first end **210** and fluidly connects to the respective pipe inlet **218** of each of the coolant pipes **216**. The second portion **238** of the manifold **222** is disposed at the second end **212** and fluidly connects to the respective pipe outlet **220** of each of the coolant pipes **216**. In this configuration, each of the coolant pipes **216** extends from the first portion **234** of the manifold **222** to the second portion **238** of the manifold **222**. The coolant supply **224** fluidly connects to the manifold inlet **236** and to the manifold outlet **240**. The coolant **226** flows from the coolant supply **224** to the manifold inlet **236**, into each pipe inlet **218**, through each coolant pipe **216**, out through each pipe outlet **220**, to the manifold outlet **240**, and back to the coolant supply **224**.

[0043] Now referring to FIG. **5**, another exemplary embodiment of the stator **250** is shown. The stator **250** of FIG. **5** may be configured in substantially the same manner as the exemplary stator of FIG. **2**, and common structures will share a common numeral and a common function as described above. In this form, a plurality of coolant pipes **252** extend along the wedges **230** of the stator core **206** beneath the plurality of wires **214**, i.e., inward in the radial direction R'. The arrangement of the coolant pipes **252** along the wedges **230** provides additional space within the slots **208** for the wires **214**. The thermally conductive potting material **228** absorbs heat from the manifold **222** and the coolant pipes **252**, further increasing heat transfer from the wires **214**.

[0044] With reference to FIG. **6**, providing a partial schematic view of the stator **250** without the thermally conductive potting material **228**, the coolant pipes **252** extend along the wedges **230** to contact the wires **214**, cooling the wires **214** with the coolant **226** flowing therethrough. In the embodiment show, the wedges **230** form a surface onto which the coolant pipes **252** mate or contact. With this mating configuration, the wedges **230** may additionally transfer heat way from the coolant pipes **252**, increasing the overall heat transfer from the wires **214**. Additionally, the wedges **230** may support the coolant pipes **252** in the slots **208**, holding the coolant pipes **252** against the wires **214** to increase contact therebetween, thereby increasing conductive heat transfer.

[0045] Now referring to FIG. **7**, providing partial schematic views of the stator **250** without the wires **214**, the coolant pipes **252** flow the coolant **226** between the first and second portions **234**, **238** of the manifold **222** in the axial direction A' to cool the wires **214**. The coolant **226** flows through the coolant pipe **252** from the pipe inlet **218** at the first end **210** to the pipe outlet **220** at the second end **212**, where the second portion **238** of the manifold **222** removes the coolant **226** to the

coolant supply **224**.

[0046] Referring now to FIG. **8**, a flow diagram of a method **300** of forming a stator of an electric machine in accordance with an exemplary aspect of the present disclosure is provided. The method **300** of FIG. **8** may be utilized to build one or more of the exemplary electric machines described above with reference to FIGS. **1-7**. Accordingly, it will be appreciated that the method **300** may generally be utilized to manufacture an electric machine for a gas turbine engine as described above. However, in other exemplary aspects, the method **300** may additionally or alternatively be utilized to manufacture any other electric machine.

[0047] As is depicted, the method **300** includes at **(302)** extending a plurality of wires through the plurality of slots along the plurality of coolant pipes. The plurality of wires extend from the first end of the stator core to the second end of the stator core.

[0048] The method includes at **(304)** extending a plurality of coolant pipes through a plurality of slots of a stator core. The coolant pipes extend in the axial direction from a first end of the stator core to a second end of the stator core. The coolant pipes may each have a pipe inlet at the first end and a pipe outlet at the second end. The coolant pipes may be disposed between adjacent windings of the plurality of wires, or the coolant pipes may be disposed beneath the plurality of wires.

[0049] The method includes at **(306)** placing a manifold around the stator core in the circumferential direction and connecting the coolant pipes to the manifold. The manifold provides coolant from a coolant supply to respective pipe inlets of each of the plurality of coolant pipes. A first portion of the manifold is placed at the first end, and a second portion of the manifold is placed at the second end.

[0050] The method includes at **(308)** applying a thermally conductive potting material to the first and second ends of the stator core, the plurality of wires, the plurality of coolant tubes, and the manifold. The thermally conductive potting material is applied as a liquid or a gel to cover the surfaces of the stator core and the manifold.

[0051] The method includes at **(310)** curing the thermally conductive potting material into a solid. Upon curing into a solid, the thermally conductive potting material secures the plurality of wires and the plurality of coolant pipes in the plurality of slots. The solid thermally conductive potting material transfers heat away from the coolant pipes and the plurality of wires, improving cooling of the stator.

[0052] The method includes at **(312)** flowing coolant through the manifold and the coolant pipes. The manifold is connected to a coolant supply that provides coolant to the coolant pipes. The coolant flows from the pipe inlet of each of the coolant pipes to the pipe outlet of each of the plurality of coolant pipes, transferring heat away from the plurality of wires.

[0053] Further aspects are provided by the subject matter of the following clauses:

[0054] A stator for an electric machine, the stator defining an axial direction, a circumferential direction, and a radial direction, the stator including a stator core defining a slot extending in the axial direction from a first end to a second end, a wire disposed in the slot, a coolant pipe disposed in the slot and extending in the axial direction from the first end to the second end, the coolant pipe extending from a pipe inlet to a pipe outlet, and a thermally conductive potting material disposed in the slot between the coolant pipe, the wire, and the stator core.

[0055] The stator of any of the previous clauses, wherein the wire is supported in the slot by the thermally conductive potting material.

[0056] The stator of any of the previous clauses, wherein the pipe inlet of the coolant pipe is disposed at the first end of the stator core, and wherein the pipe outlet of the coolant pipe is disposed at the second end of the stator core.

[0057] The stator of any of the previous clauses, wherein the thermally conductive potting material secures the wire to the stator core.

[0058] The stator of any of the previous clauses, wherein the thermally conductive potting material is an epoxy resin, a silicon-based resin, or both.

[0059] The stator of any of the previous clauses, wherein the coolant pipe contacts the wire.

[0060] The stator of any of the previous clauses, further including a coolant disposed in the coolant pipe, the coolant configured to dissipate heat from the wire upon flowing through the coolant pipe.

[0061] The stator of any of the previous clauses, further including a manifold extending around the stator core in the circumferential direction, the manifold fluidly connected to the coolant pipe.

[0062] The stator of any of the previous clauses, wherein the manifold is connected to the pipe inlet of the coolant pipe and connected to the pipe outlet of the coolant pipe.

[0063] The stator of any of the previous clauses, wherein the thermally conductive potting material is disposed on the manifold.

[0064] The stator of any of the previous clauses, further including a plurality of coolant pipes including the coolant pipe, wherein at least some of the plurality of coolant pipes are arranged in parallel to each other of the at least some of the plurality of coolant pipes.

[0065] The stator of any of the previous clauses, further comprising a coolant supply fluidly connected to the coolant pipe.

[0066] The stator of any of the previous clauses, wherein the stator core defines a plurality of slots.

[0067] The stator of any of the previous clauses, further comprising a plurality of wires.

[0068] A method of forming a stator of an electric machine, the stator defining an axial direction and a circumferential direction, the method including extending a plurality of wires in each of a plurality of slots of a stator core, extending, in each of the plurality of slots, at least one of a plurality of coolant pipes in the axial direction, and applying a thermally conductive potting material to an end of the stator core over the plurality of wires and the plurality of coolant pipes.

[0069] The method of any of the previous clauses, wherein the thermally conductive potting material is an epoxy resin, a silicon-based resin, or both.

[0070] The method of any of the previous clauses, further including curing the thermally conductive potting material to secure the plurality of wires to the stator core.

[0071] The method of any of the previous clauses, further including flowing a coolant through the plurality of coolant pipes to dissipate heat from the plurality of wires.

[0072] The method of any of the previous clauses, further including extending the at least one of the plurality of coolant pipes along a wedge of the stator core.

[0073] The method of any of the previous clauses, further including placing a manifold around the stator core in the circumferential direction, the manifold in fluid communication with the plurality of coolant pipes, and, then, applying the thermally conductive potting material to the end of the stator core and over the manifold.

[0074] The method of any of the previous clauses, wherein each of the plurality of coolant pipes extends from a pipe inlet connected to the manifold to a pipe outlet connected to the manifold.

[0075] An aeronautical gas turbine engine includes a turbomachine including a compressor, a combustor, and a turbine arranged in serial flow order and an electric machine defining an axial direction, the electric machine including a stator and a rotor, the rotor rotatable with the turbomachine, the stator including a stator core defining a plurality of slots extending in the axial direction from a first end to a second end, a plurality of wires disposed in the plurality of slots, a plurality of coolant pipes disposed through the plurality of slots and extending in the axial direction from the first end to the second end, each of the plurality of coolant pipes extending from a pipe inlet to a pipe outlet, and a thermally conductive potting material disposed in the plurality of slots between the plurality of coolant pipes, the plurality of wires, and the stator core.

[0076] The aeronautical gas turbine engine of any of the previous clauses, wherein the plurality of wires are supported in the slot by the thermally conductive potting material.

[0077] The aeronautical gas turbine engine of any of the previous clauses, wherein the pipe inlet of at least one of the plurality of coolant pipes is disposed at the first end of the stator core, and wherein the pipe outlet of at least one of the plurality of coolant pipes is disposed at the second end of the stator core.

[0078] The aeronautical gas turbine engine of any of the previous clauses, wherein the thermally conductive potting material secures the plurality of wires to the stator core.

[0079] The aeronautical gas turbine engine of any of the previous clauses, wherein the thermally conductive potting material is an epoxy resin, a silicon-based resin, or both.

[0080] The aeronautical gas turbine engine of any of the previous clauses, wherein at least one of the plurality of coolant pipes contacts at least one of the plurality of wires.

[0081] The aeronautical gas turbine engine of any of the previous clauses, further including a coolant disposed in the plurality of coolant pipes, the coolant configured to dissipate heat from the plurality of wires upon flowing through the plurality of coolant pipes.

[0082] The aeronautical gas turbine engine of any of the previous clauses, further including a manifold extending around the stator core in the circumferential direction, the manifold fluidly connected to the plurality of coolant pipes.

[0083] The aeronautical gas turbine engine of any of the previous clauses, wherein the manifold is connected to the pipe inlet of at least one of the plurality of coolant pipes and connected to the pipe outlet of at least one of the plurality of coolant pipes.

[0084] The aeronautical gas turbine engine of any of the previous clauses, wherein the thermally conductive potting material is disposed on the manifold.

[0085] The aeronautical gas turbine engine of any of the previous clauses, wherein at least some of the plurality of coolant pipes are arranged in parallel to each other of the at least some of the plurality of coolant pipes.

[0086] The aeronautical gas turbine engine of any of the previous clauses, further comprising a coolant supply fluidly connected to at least one of the plurality of coolant pipes.

[0087] This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A stator for an electric machine, the stator defining an axial direction, a circumferential direction, and a radial direction, the stator comprising: a stator core defining a slot extending in the axial direction from a first end to a second end; a wire disposed in the slot; a coolant pipe disposed in the slot and extending in the axial direction from the first end to the second end, the coolant pipe extending from a pipe inlet to a pipe outlet; and a thermally conductive potting material disposed in the slot between the coolant pipe, the wire, and the stator core.

2. The stator of claim 1, wherein the wire is supported in the slot by the thermally conductive potting material.

3. The stator of claim 1, wherein the pipe inlet of the coolant pipe is disposed at the first end of the stator core, and wherein the pipe outlet of the coolant pipe is disposed at the second end of the stator core.

4. The stator of claim 1, wherein the thermally conductive potting material secures the wire to the stator core.

5. The stator of claim 1, wherein the thermally conductive potting material is an epoxy resin, a silicon-based resin, or both.

6. The stator of claim 1, wherein the coolant pipe contacts the wire.

7. The stator of claim 1, further comprising a coolant disposed in the coolant pipe, the coolant

configured to dissipate heat from the wire upon flowing through the coolant pipe.

8. The stator of claim 1, further comprising a manifold extending around the stator core in the circumferential direction, the manifold fluidly connected to the coolant pipe.

9. The stator of claim 8, wherein the manifold is connected to the pipe inlet of the coolant pipe and connected to the pipe outlet of the coolant pipe.

10. The stator of claim 8, wherein the thermally conductive potting material is disposed on the manifold.

11. The stator of claim 1, further comprising a plurality of coolant pipes including the coolant pipe, wherein at least some of the plurality of coolant pipes are arranged in parallel to each other of the at least some of the plurality of coolant pipes.

12. The stator of claim 1, further comprising a coolant supply fluidly connected to the coolant pipe.

13. A method of forming a stator of an electric machine, the stator defining an axial direction and a circumferential direction, the method comprising: extending a plurality of wires in at least one of a plurality of slots of a stator core; extending, in at least one of the plurality of slots, at least one of a plurality of coolant pipes in the axial direction; and applying a thermally conductive potting material to an end of the stator core over the plurality of wires and the plurality of coolant pipes.

14. The method of claim 13, wherein the thermally conductive potting material is an epoxy resin, a silicon-based resin, or both.

15. The method of claim 13, further comprising curing the thermally conductive potting material to secure the plurality of wires to the stator core.

16. The method of claim 13, further comprising flowing a coolant through the plurality of coolant pipes to dissipate heat from the plurality of wires.

17. The method of claim 13, further comprising extending the at least one of the plurality of coolant pipes along a wedge of the stator core.

18. The method of claim 13, further comprising placing a manifold around the stator core in the circumferential direction, the manifold in fluid communication with the plurality of coolant pipes, and, then, applying the thermally conductive potting material to the end of the stator core and over the manifold.

19. The method of claim 18, wherein at least one of the plurality of coolant pipes extends from a pipe inlet connected to the manifold to a pipe outlet connected to the manifold.

20. An aeronautical gas turbine engine comprising: a turbomachine comprising a compressor, a combustor, and a turbine arranged in serial flow order; and an electric machine defining an axial direction, the electric machine comprising a stator and a rotor, the rotor rotatable with the turbomachine, the stator comprising: a stator core defining a plurality of slots extending in the axial direction from a first end to a second end; a plurality of wires disposed in the plurality of slots; a plurality of coolant pipes disposed in the plurality of slots and extending in the axial direction from the first end to the second end, at least one of the plurality of coolant pipes extending from a pipe inlet to a pipe outlet; and a thermally conductive potting material disposed in the plurality of slots between the plurality of coolant pipes, the plurality of wires, and the stator core.
